

The Price of Thought

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Takeaways

- The cost of AI performance has been falling 39–52% per quarter in the last 3 years, the speed of the fall depending on the performance benchmarks. These figures assume one is using the cheapest model up to the task at any given time.
- Across the five benchmarks for which we have the best data—testing knowledge in reasoning ability in mathematics, hard sciences, chess, and other games—the average decline rate is 47%/quarter. That is equivalent to a 12-fold drop every 12 months.
- Often the cost of a given level of performance falls fastest right after it is first achieved, that is, when it is state-of-the-art (SOTA). Across our five “primary” benchmarks, three exhibit this pattern. Averaging across all five, cost falls 67%/quarter for performance that has just debuted as SOTA. Two years later, prices fall half as fast, at a still-brisk 32%/quarter.
- If this pace continues for years—a big “if”—it will far outstrip the declines seen for other transformative technologies. The price of thought is currently falling 3.7 times faster than did the price of compute in the 1980s and 1990s, 17 times faster than the price of lithium batteries since 1991, and 67 times faster than the price of lighting since 1800.

The figure below foregrounds select data points in order to convey patterns in the full body of data. For each of the primary benchmarks, it shows the evolution over time of the cost of achieving 10% performance, or 50% performance, or 90% performance.¹ Each data point marks the arrival of a large language model (LLM) that sets a new cost record for that level of performance. The markers along each cost trace are shaded according to the corresponding level on the [Epoch Capability Index](#), an overall indicator of LLM strength. The vertical axis, for cost, is logarithmic, so that straight lines correspond to steady declines in percentage terms, such as 50%/quarter. When the cost is already a tenth of a penny, a 50% further drop, though dramatic, amounts to little in absolute terms. But we see that even this *relative* rate of decline tends to slow.

¹ As explained below, the performance scales are adjusted for guessing. For example, the GPQA Diamond exam consists of multiple-choice questions with four choices. Just by guessing, an AI would get a quarter of the questions right. So for this benchmark, 10% performance means the level 10% of the way from 25% to 100%.

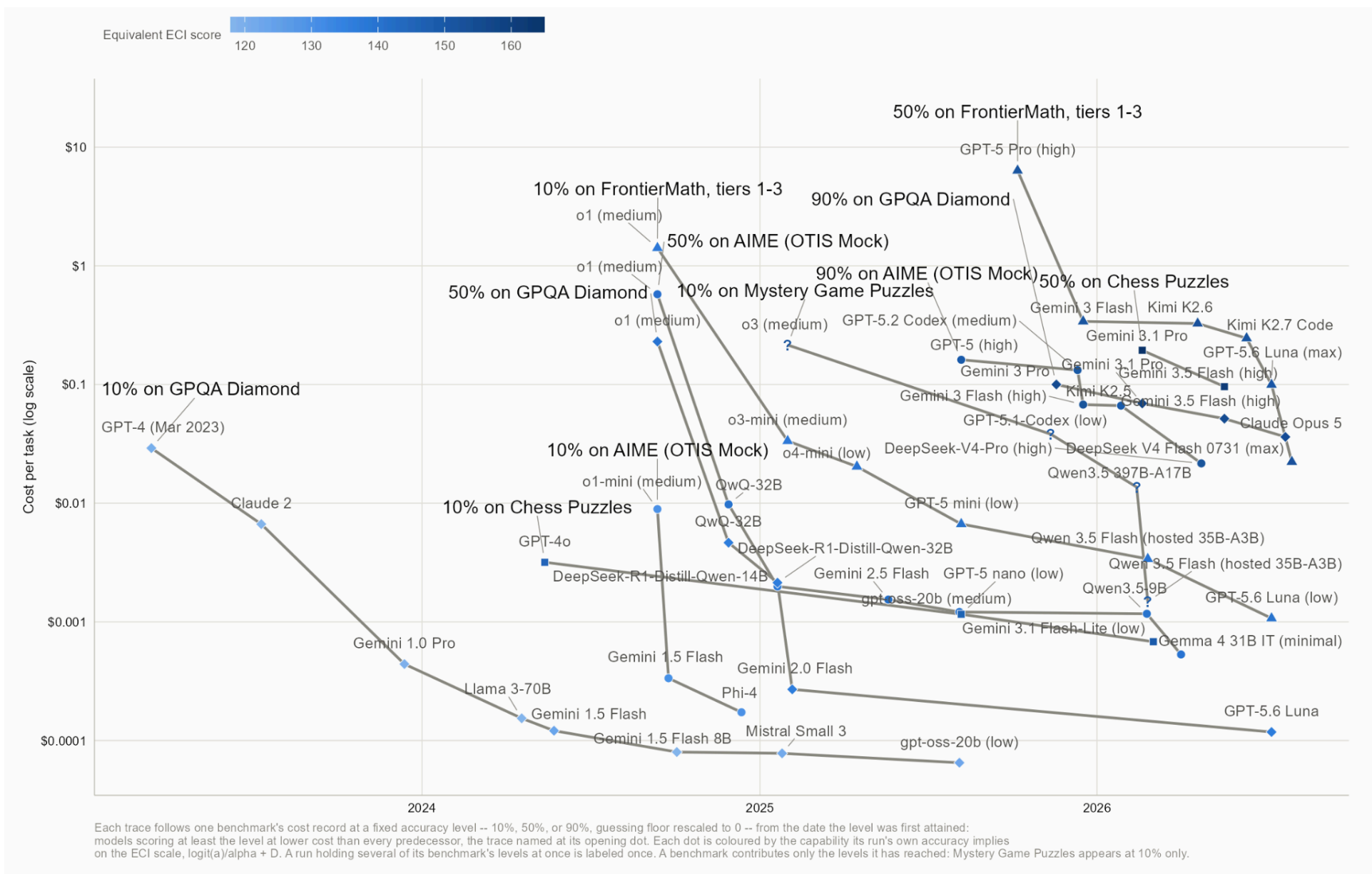
With prices collapsing, it might appear that AI tokens will soon be too cheap to meter. But that intuition is wrong, for two reasons. First, if an important job for AI, like analyzing thousands of MRIs to study disease processes in the brain, is equivalent to running one of these benchmark tests a billion times, then it remains economically consequential whether the price of this performance is \$0.01 or \$0.001 per run. Second, even as prices fall, AI's capability rises, allowing it to tackle new classes of problems. The price of passing a first-grade math test may now be trivial. The price of proving theorems is not.

The repo is [here](#). GitHub pages are built from it; go [here](#) to explore many plots and tables.

Limitations

[JS can do]

Figure 1. Cost records at 10%, 50%, and 90% accuracy on primary benchmarks



Data

When attempting to measure tradeoffs between cost and performance, an immediate concern is that individual benchmark scores only convey information about a single point along the cost-performance continuum. Previous literature has highlighted the importance of inference time scaling for benchmark elicitation – using more tokens can produce better scores, but at a higher cost. The concern goes both ways; cost-sensitive users may choose to run models with minimal budgets, accepting a lower level of performance. At a first glance, this appears to mean that we’d need to run multiple evaluations in order to characterize the full range of tradeoffs.

[Researchers from CAISI document one approach to handling this ...]

As used here, the data set adjusts for guessing. Under the CAISI methodology for estimating performance under budget constraints, models are first run without budget limit. Then to estimate performance at various budget levels, they are scored as if they completed a subset of questions. Here, unanswered questions, instead of being coded as 0, are coded with the probability of guessing the correct answer. So, for example, in the GPQA Diamond benchmark, whose questions are four-way multiple-choice, if a model only answers one question before running out of budget, the rest are scored as if (in expectation) the model guessed on the rest. Then all GPQA scores are clipped to 0.25 and rescaled from the [0.25, 1] range to [0, 1].

Modeling

The challenge

For each benchmark, we have a set of triples (a_i, t_i, c_i) representing the accuracy of a model’s answers (between 0 and 1), the model’s release date, and the dollar cost of the run. Tests are typically repeated while varying settings such as the level of reasoning effort, so we can have more than one triplet for any given model, benchmark, and cost level.

We want to characterize the rates of price decline for given levels of performance—not of the average model, but of the “frontier” model, meaning the cheapest available that can deliver the required performance on a benchmark.

To be concrete, we define two empirical Pareto frontiers, for accuracy and cost. The empirical accuracy frontier is the highest performance achieved within a given budget using models available up to a given date:

$$A(t, c) = \max_{i|t_i \leq t, c_i \leq c} a_i$$

The empirical cost frontier is the lowest cost at which at least one available model can meet or beat a given accuracy target:

$$C(t, a) = \min_{i|t_i \leq t, a_i \geq a} c_i$$

These surfaces have a bi-directional staircase character, as they experience discontinuous jumps in either of their dimensions, always in the same direction.

Representing these moving frontiers with a statistical modeling problem is challenging in a few ways:

1. Output is bounded above, since accuracy can't exceed 100%. In most economic models of production and cost frontiers, output can go arbitrarily high.
2. The frontier is constructed as a maximum or minimum at each point. This makes it unstable. A single benchmark result—or a single missing one—can substantially shift the frontier for months. Most statistical work focuses on averages. In this setting, each additional data point tends to affect results less than the previous one, as the estimate of the average converges. When studying extrema, this limiting behavior may either not set in, or set in more slowly. So we should presume that the empirical frontiers are noisy representations of “true” frontiers.
3. Because of #2, the sampling program matters more. E.g., if a higher percentage of newer models are benchmarked, this will tend to puff out the later regions of the frontiers. Because testing models takes time and money, we have mostly tested models that appear to be near the frontier as of their release date. The sampling is therefore a bit ad hoc. That means that we do not have a credible, clean model of the data generating process, in a context where it might substantially affect the results.
4. Within the time frame studied, change is extremely fast. Most studies of production frontiers take place in less dynamic industries, like manufacturing and farming, where one can imagine a large number of producers tracing out the frontier at any given time.

Items 2 and 3 together mean that when we model the frontier, it's hard to produce credible standard errors, thus estimates of statistical significance. Those are always founded on a model of the data generating process that looks roughly realistic, including in its specification of noise components.

The models

We fit five main models to the data, two to the empirical frontiers and three to the full set of benchmarking results. Of the latter, two are applications of stochastic frontier analysis. Their results are not credible and are relegated to the appendix.

Three variables in play: release date, spending, and benchmark performance. It is natural to model the last as a function of the other two: spending is an input to production, time affects the technology of production, and performance is the output. But our ultimate interest here is in determinants of cost, not performance. And there is a long tradition in economics of viewing production and cost functions as duals, and modeling both with similar methods. The dual perspective situates cost as a function of time and accuracy.

Ideally the implications for the rates of cost decline should not depend on whether we model performance as a function of time and cost or cost as a function of time and performance. Either way we are estimating the same relationships among variables. And once we have fit a model to one frontier we can back out the implications for the other. In practice, there are some asymmetries (or anti-symmetries) between how the same modeling approach is applied to the two frontiers. Consider: in fitting a line to a cloud of points in the plane, one could minimize the vertical distances between the points and the line or the horizontal distances, and thus get somewhat different answers.

The approach here is to model from both perspectives, and then from both draw out the implications for model performance and cost. The [web-based interface](#) makes the results browsable.

Model 1: Pareto logistic/least squares

The idea is to fit smooth surfaces to the empirical frontiers defined above. Like best-fit lines through a cloud of dots, the best fits will have real data points on both sides. Thus the fitted frontiers will not technically embrace every point that contributes to the empirical frontiers, but will serve to summarize their overall shapes.

For the *accuracy* model, we assume that at any given time, frontier model accuracy is related to both log cost and release date according to an S curve:.

$$E(a|x_a) = \sigma(x_a' \beta_a)$$

σ is the logistic S curve function, which rises from 0 in the $-\infty$ direction to 1 at $+\infty$. x_a is a vector of controls—at a minimum, log cost ($\ln c$), release date (t), and a constant. β_a is the corresponding vector of coefficients. $E(a|x_a)$ is the expected accuracy of a model given values for those controls.

The parameters are estimated with a “fractional logit” regression ([Papke and Wooldridge 1996](#)), whose maximand has the same mathematical formula as the logit model for binary outcomes, and is reinterpreted to allow outcome values between 0 and 1, inclusive.

For the *cost* frontier, we model log cost, which is unbounded, as linear in its predictors:

$$E(\ln c|x_c) = x_c' \beta_c$$

x_c contains, at a minimum, the logit/inverse logistic of a , symbolized $\sigma^{-1}(a)$, along with t and a constant. This is fit with ordinary least squares. Because the logit transform maps 0 and 1 to $\pm\infty$, unlike in the previous model, scores of 0 or 1 are dropped from the sample. That includes cases when a model scores at or below the expected score from guessing. For example, if an LLM scores 0.25 on GPQA Diamond, the rescaling of the score range from $[0.25, 1]$ to $[0, 1]$ will give that model a guess-adjusted score of 0.

These models for accuracy and cost are each fit once, across their two-dimensional domains. In the fits, the frontier surface is represented not just by the record-setting data points of the sort highlighted in Figure 1, but by the levels of $A(t, c)$ or $C(t, a)$ over a grid of points. The grids’ points are evenly spaced values of 1) release date and 2) log cost or accuracy.² In the time dimension, the grids have a resolution of 1 point about every 13 days, chosen so that there are 100 points for the benchmark whose data starts first (AIME OTIS Mock, applied to GPT-4, which was released on March 14, 2023). And the grids have 100 points in the 0–1 accuracy range. The use of a grid reflects the idea that the absence of a new cost record on any given date—the absence of a data point—is itself information. If standard errors were reported for these regressions, then this gridding could be problematic since arbitrarily raising the resolution of the grids would increase the sample size, and thus, naively, reduce standard errors.³

Because entire Pareto *surfaces* are being modeled, any particular slice of the best-fit surface, such as for accuracy versus cost as of release date 2026-01-01, will generally *not* be the best fit just for the corresponding slice of the empirical frontier. The quality of fit in any slice is constrained by the goal of finding the best overall fit across all slices, subject to the structure of the model.

Model 2: Envelope

This model is the same as the previous one except that

² It bears emphasizing that the way the grid is constructed affects results because it effectively weights certain regions of data space relative to each other. For example, In the accuracy dimension, the grid could be evenly spaced in logit accuracy rather than accuracy. This would put many points, thus more weight, very close to 0. But we are not especially interested in low-accuracy results.

³ This problem might be handled with methods for spatially autocorrelated data.

- It forces the theoretical frontier to lie completely outside the empirical frontier—above when modeling accuracy and below when modeling cost.
- In the accuracy model, some benchmark tests produce perfect scores of 1. But it is impossible for an S-curve-based envelope to ever reach 1, thus for the envelope to embrace such a point of perfection. So before fitting, 1's are replaced with $1 - 1/2n$ where n is the number of questions a model was challenged to answer in forming the data point, i.e, the number of runs times number of questions per run (column "n_samples" in [cost truncated curves.csv](#)). In other words, perfect scores are replaced with a number that represents getting all questions right half the time and all but one right the other half.

The requirement that the envelopes stay outside all of the corners of the empirical frontier staircases (or just touching them) constitutes a large set of nonlinear constraints on the parameters. Sequential Least Squares Programming (SLSQP) is used to find the constrained best fits.

Model 3: Full-data logistic/least squares

This model is like model 1 except that it is fit to all the test results in the data set, rather than a grid of points representing the empirical frontiers. While this model is not structurally focused on the frontier, the fits are included for a few reasons. They are not as intrinsically unstable since they are not based on the min and max operators. The approach is simple. And they might be expected to give about the same answers as models of the frontiers. As an analogy, we would expect that the rate of decline in computer memory prices over the last 50 years is about the same whether or not one explicitly models the empirical frontier. Here, of course, the time span is much shorter and the progress faster.

Nonlinear models

The models just defined are linear in the sense that the vector x consists solely of two predictors, namely time, and accuracy or log cost. (The accuracy model remains nonlinear in the sense that it incorporates the logistic transform.) For the accuracy models, this means assuming that the passage of each quarter, and each doubling of spending, causes the same improvement in the logit of benchmark accuracy, $\sigma^{-1}(a)$. For cost models, it assumes that the passage of each quarter, and each one-unit increase in the logit of accuracy, has the same proportional impact on cost.

These assumptions are not obviously true. Cost declines might accelerate or decelerate in calendar time. There might be diminishing returns to spending at any given time.

To allow for nonlinearities, all models are also run in quadratic variants. In the accuracy models, the vector x gains t^2 , $(\ln c)^2$, $t \times \ln c$. In the cost models, it gains t^2 , $(\sigma^{-1}(a))^2$, $t \times \sigma^{-1}(a)$.

While mathematically natural, a peculiarity of quadratic models is that they can imply parabolic trends, with directional changes. LLMs then are predicted to get worse over time at certain times or cost levels. Ordinarily these implausible projections appear outside or on the fringes of the data range. But they can still produce model fits that are hard to explain.

An alternative approach is apply the Box-Cox transform to each of the x variables. This transform has the greater virtue of offering a range of curvatures while avoiding directional changes, i.e., preserving monotonicity. For example, for cost, we would apply

$$\phi_{\lambda_c}(c) = \frac{c^{\lambda_c} - 1}{\lambda_c}$$

This introduces the exponent λ_c as an additional parameter to be estimated. The Box-Cox transform shifts and scales the power transform in order to give it some nice properties. When $\lambda_c = 1$, it is just $\phi_{\lambda_c}(c) = c$, so it includes linearity as a special case. When $\lambda_c > 1$ it is convex and when $\lambda_c < 1$, concave. In the $\lambda_c \rightarrow 0$ limit, it equals $\ln c$.

In the models with Box-Cox transforms, each of the predictor variables is allowed its own λ exponent. When brought to the accuracy variable, the transform is applied to the odds ratio $a/(1 - a)$. That way the transform embraces $\sigma^{-1}(a) = \ln[a/(1 - a)]$ as a special case, which is used in the linear cost model. In addition, just as in the quadratic models, the *product* of the two Box-Cox-transformed terms is also entered as a regressor. Here, the same λ exponents are carried over.

Under the Box-Cox transform, the possible values of $\phi_{\lambda_t}(t)$ depend on where “time zero” is. We define it as November 18, 2021, which is when OpenAI [fully opened the GPT-3 API for commercial use](#). first started charging for access to GPT-3. A different choice would cause the results to shift somewhat.

A complication in terminology: [Box and Cox \(1964\)](#) applied the “Box-Cox transform” to dependent variables, to make their distributions more normal-like. Here we are following [Box and Tidwell \(1962\)](#) in applying it to the independent variables. So these regressions are more probably labeled as Box-Tidwell.

The fitting code conducts the search for the best fit with a two-level process. In the upper level, a search is conducted in the space of possible λ pairs. For each set of trial values, the other model parameters are estimated using the same methods as for the linear models while holding the λ exponents fixed.

The exponents on time, λ_t , sometimes proved hard to estimate, especially in the cost models for benchmarks with short testing histories. No peak would be found within the (wide) allowed range of $[-8, 8]$. This is not surprising since the time spans of the data sets for each benchmark are so short, making it hard to detect 2nd-order effects in time. When estimation fails in this way, the code imposes $\lambda_t = 1$, reverting to linearity.

Preferences among the models

Because of our interest in cost declines at the frontier, we favor the models that are fit to that frontier rather than the full data set, and to the cost frontier in particular. The stochastic frontier models described in the appendix fail to credibly identify a (moving) frontier. Simple regression on the full data set (model 3 above) is not designed to identify the frontier.

Of the two models fit just to the frontier, we prefer model 1, which is not constrained to envelope the empirical Pareto surface. Our interest is not in where some theoretical frontier lies, beyond the reach of most models at any given point in time, but in real-world cost-performance tradeoffs. Our data are representative of available performance because they cover essentially all the important models. A smooth approximation to the empirical frontier surface distills its angular contours into a few numbers.

As for the order of the model, we prefer Box-Tidwell. In the [3-D surface plots](#), the linear fits often look unrealistic. Quadratic models do better, but often with unrealistic reversals at the edges of the data range. The Box-Tidwell fits follow the data more closely while looking more realistic because of the monotonicity.

Figures 2 and 3 are a sample of the graphical presentations of the preferred model fits. In Figure 2, data points for four of the five primary benchmarks are plotted, with color indicating the LLM's release date. Superimposed are semiannual accuracy frontiers, and the corresponding smooth curves implied by the model fits. Figure 3 switches to modeling cost, and compares the full Pareto surfaces to their Box-Tidwell fits. (Note the direction of the time axes.) The live plots are interactive. You can zoom and rotate them to gain perspective.

Figure 2. Benchmark results, semiannual Pareto accuracy frontiers, and smooth fits thereto, over time, four primary benchmarks

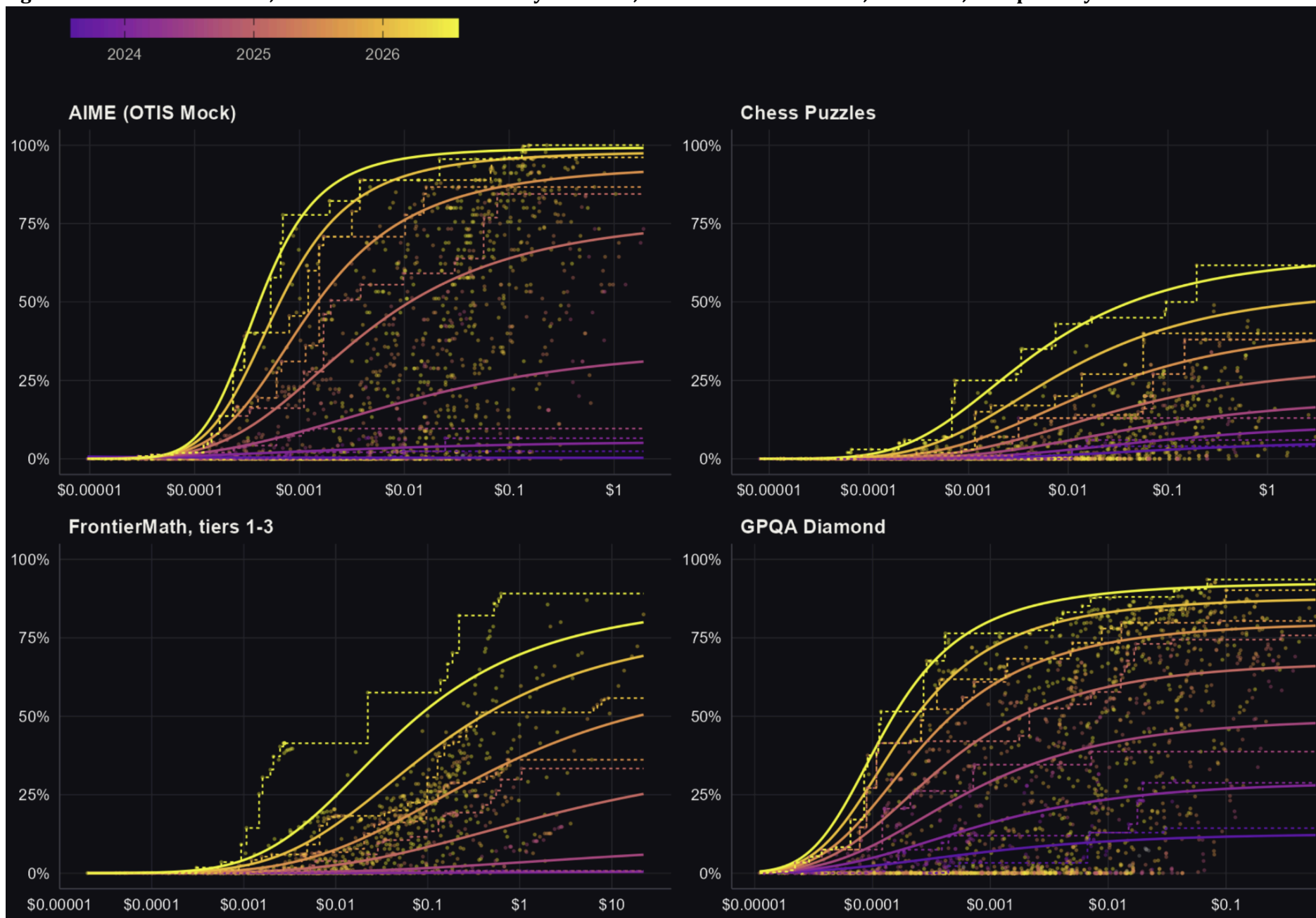
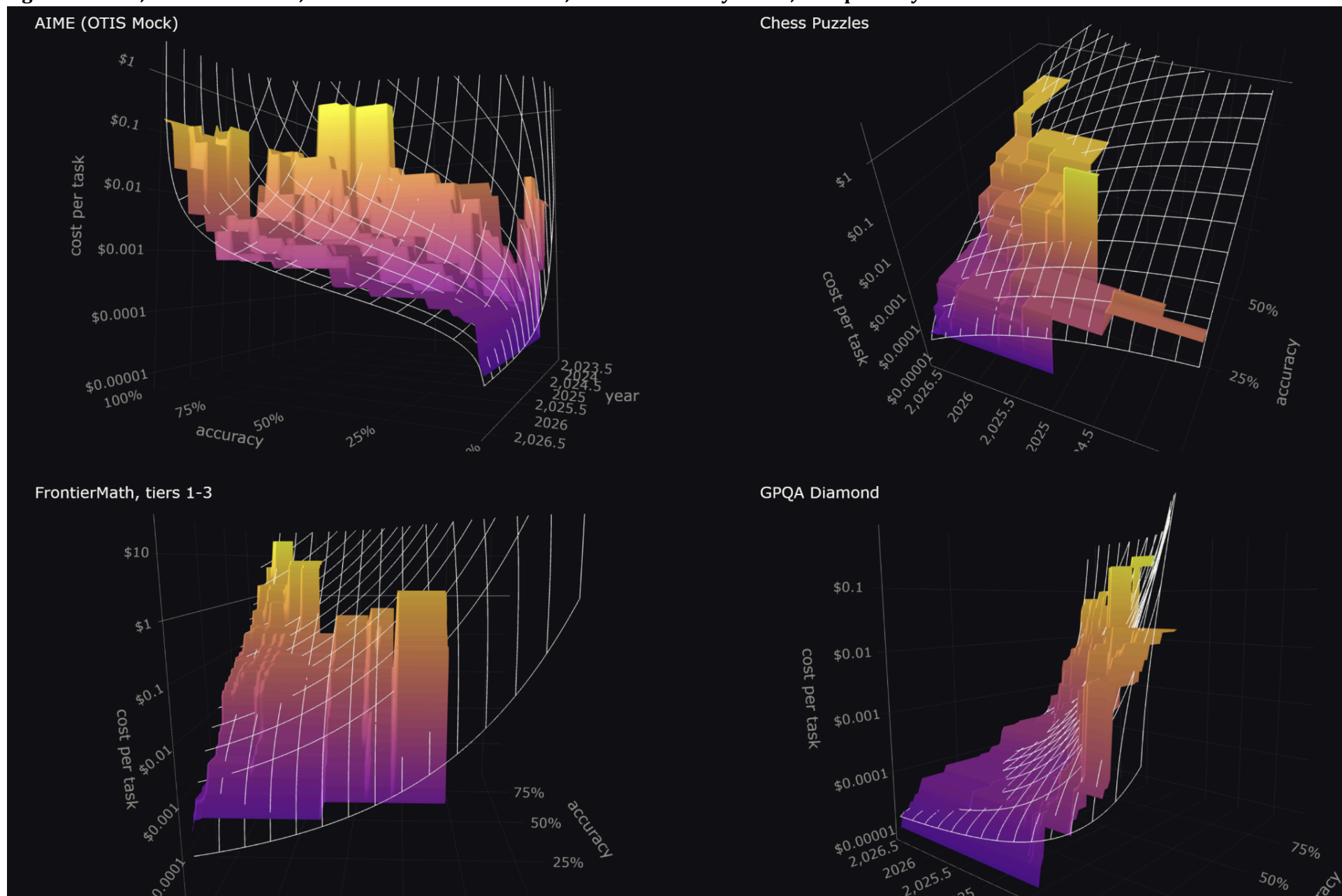


Figure 3. Costs, Pareto frontiers, and smooth fits to frontiers, various accuracy levels, four primary benchmarks



Extracting estimates of cost declines

The linear cost models directly provide estimates of the impact of time (release date) on log cost, in the form of a coefficient β_t .

To extract the equivalent from linear accuracy models takes more work. The model fits give rise to isoquants, curves in the cost-time plane along which predicted accuracy is constant. Movement along these iso-accuracy curves is characterized by

$$da = \frac{\partial a}{\partial t} dt + \frac{\partial a}{\partial \ln c} d \ln c = 0$$

Solving for $d \ln c / dt$,

$$\frac{d \ln c}{dt} = - \frac{\frac{\partial a}{\partial t}}{\frac{\partial a}{\partial \ln c}} = - \frac{\beta_t}{\beta_{\ln c}}$$

where the β 's are coefficients from an accuracy model fit.

Extrapolating individual benchmark results to the ECI

The [Epoch Capabilities Index](#) integrates thousands of benchmark results for hundreds of AI models in order to estimate each model's overall capability. It [is built](#) on an equation that may be written

$$\sigma^{-1}(a_{m,b}) = \alpha_b (C_m - D_b)$$

$a_{m,b}$ is the accuracy of model m on benchmark b , C_m is the capability of model m , normalized so that Sonnet 3.5 gets a 130 and GPT-5 a 150. D_b is the difficulty of a benchmark, on the same scale. The “discrimination” parameter α_b is a benchmark-specific conversion rate from ECI capability units to logits of performance on the benchmark.

This model comes out of Item Response Theory, which is the dominant method for assessing the difficulty of standardized tests and scoring students who take them. Here, though individual benchmark results rather than individual questions are the unit of observation. The core idea is that how well a model performs on a benchmark depends only on the difference between its capability, C_m , and the benchmark's difficulty, D_b . The α_b values determine how sensitive the expected score is to that difference, i.e., how quickly a series of ever-better models will climb the S-curve from poor to excellent performance.. If

α_b is low, then it takes a lot of capability improvement to climb the curve, but there is a wide range of capability levels over which further improvement will noticeably raise scores. On the other hand, if α_b is high, then once models approach a certain capability threshold ($C_m \approx D_b$), they will cross rapidly from abysmal to excellent performance. Soon after, the benchmark will be saturated, and further increases in capability will be hard to detect with this benchmark. High discrimination occurs when the questions on a test demand nearly the same skills.

Results

Average rates of decline

We focus first on estimates of the overall rate of cost decline. This is naturally of interest. In addition, it allows us to validate model results against model-free estimates.

We stress that there are in fact many averages one could compute. The differences come down mostly to how much to weight different regions of data space. Should experience in the \$1–2 range be granted the same importance as that in \$3–4, or should we equal-weight across orders of magnitude, so that decline episodes in the \$0.0001–0.001 range matter as much as those \$1–10? Should the benchmarks be weighted equally, or should those with longer history be given more weight? Random-effects meta-analysis, which is a workhorse in evidence synthesis, tends toward equal weighting when individual “studies” produce precise but statistically incompatible results, which point to different effects in different study settings; but here the ECI paradigm implies that capability is unitary, and anyway our preferred estimates lack standard errors to indicate precision. And so on.

Our preferred approach arises from an attempt to balance parsimony, transparency, and conceptual coherence.

Table 1 shows many estimates of the average quarterly rate of cost decline. The first result column shows the average quarterly drop in the cost of frontier performance estimated using *none* of the foregoing models. Here, model-free estimates are made by averaging empirical cost declines over a large set of points in data space. In particular, the code creates grids in time and accuracy identical to those described earlier for the model fitting. At each grid point (t, a) , and for each benchmark, we compute the empirical frontier cost decline ratio over the following quarter, $C(t + 0.25, a)/C(t, a)$. Each benchmark’s ratios are then averaged geometrically. And then subtracted from 100%.

Among the five primary benchmarks, the decline rates are fairly close together, ranging from 43%/quarter for Chess Puzzles to 53%/quarter for FrontierMath, tiers 1–3. Their simple average is 47%/quarter.

The following columns show results from the cost models described above, in their linear and Box-Tidwell variants. The models, as models, have the disadvantage of imposing a certain mathematical structure, which is never exactly correct. On the other hand, to the extent the models *are* correct, their results will be more precise.⁴

In the second column, we see that fitting a linear model to the empirical log cost frontier—linear, that is, in t and $\sigma^{-1}(a)$ —produces results that are reasonably similar to the model-free ones. The largest disagreement is about 11 points, for the science benchmark GPQA Diamond. Giving the model more flexibility—with Box-Cox transforms of time and the accuracy odds ratio, along with their product—generally improves the concordance with the model-free estimates for the five primary benchmarks (next column of the table). Because this model is nonlinear, the rate of cost decline depends on time and accuracy, so no one parameter constitutes an overall bottom-line on the rate of cost decline. To distill the model’s overall implications, we average over the same grid of cost and accuracy values as before. This time what is averaged is the model’s predictions for the instantaneous rate of cost decline at each grid point. Except in the Mystery Game Puzzles, the overall decline rate moves closer to the model-free number. (Mystery Game Puzzles are like chess problems, except in a different game, whose identity is kept confidential, to prevent AI companies from deliberately training for the test.) **For the five primary benchmarks, the average decline rate is 47.0%/quarter**, an exact match to the model-free average.

Fits of the other models deviate substantially from those just covered. Requiring the predicted cost frontier to perfectly envelope the empirical frontier produces some implausible values, like 2.7%/quarter (linear) and 7.6%/quarter (Box-Tidwell) for SWE-Bench Verified, a measure of coding ability. Modeling *all* of the data, not just the points along the empirical cost frontier (final pair of columns), produces implausible results, including many negative signs that imply rising cost.

It is interesting to note that these cost declines are far faster than those seen for other technologies—though also potentially misleading, because the timeframe for our data set is short. Updating [Nordhaus \(1997\)](#), [Fouquet and Pearson \(2006\)](#) and [Fouquet \(2026\)](#), provide a data series on the price of a lighting—pure illumination—back to 1300. Since 1800, the price to produce a million lumen-hours has fallen 3.7%/year. Nordhaus (2007, Table 8) estimates that the price of computation fell 74.3%/year in the 1980s and

⁴ If a two-dimensional data set is indeed generated by the classical linear model—a straight line plus an independent error term—then estimating the line’s slope by fitting a model embodying that structure will be more precise than calculating a lot of slopes between various points and averaging them.

60.8%/year in the 1990s (logarithmic growth rates). Lithium batteries [dropped in price](#) from \$9,200/kilowatt-hour in 1991 to \$78 in 2024, or about 15%/year. Thus, the price of thought has fallen 67 times faster than the price of light, 17 times faster than the price of lithium batteries, and 3.7 faster than the price of computation in the 1980s and 1990s.⁵ Of course, we don't know how long AI costs will drop so fast.

⁵ All computations carried out in log points. The 46.7%/quarter average drop equates to $-4 \times \ln(1 - 0.467) \times 100 = 252$ log points/year. The price of light [fell](#) an average 3.7 log points/year, from £8497 in 1800 to £2.15 in 2023 (after inflation adjustment). The two Nordhaus figures average to 67.6 log points. The fall for batteries is $\ln(78 / 9210) / (2024 - 1991) = 14.4$ log points/year..

Table 1. Average quarterly rate of cost decline, holding accuracy fixed

			Model frontier, require envelopment					
Benchmark	Data start	Model-free	Model frontier				Model all data	
			Linear	Box-Tidwell	Linear	Box-Tidwell	Linear	Box-Tidwell
Model order								
Primary benchmarks								
AIME (OTIS Mock)	2023-03	47.1%	43.3%	50.0%	28.6%	26.1%	−1.2%	3.2%
Chess Puzzles	2023-06	43.0%	35.2%	43.4%	27.9%	46.3%	3.9%	5.9%
FrontierMath, tiers 1-3	2023-11	53.1%	50.5%	52.4%	56.6%	60.5%	24.0%	24.8%
GPQA Diamond	2023-03	47.0%	35.8%	49.3%	4.4%	26.5%	−8.4%	9.0%
Mystery Game Puzzles	2023-06	44.0%	39.0%	38.7%	4.3%	21.3%	5.0%	6.5%
Average		47.0%	41.0%	47.0%	27.2%	38.1%	5.3%	10.2%
Secondary benchmarks								
FrontierMath 2025-02	2024-05	43.3%	42.5%	48.9%	8.7%	19.6%	−8.0%	−6.4%
FrontierMath, tier 4 2025-07	2024-06	26.0%	27.8%	27.2%	0.0%	4.4%	−33.3%	−36.3%
FrontierMath, tier 4 v2	2025-01	65.1%	54.4%	76.7%	54.4%	73.4%	30.3%	49.5%
MATH, level 5	2023-06	51.8%	49.2%	57.3%	15.0%	37.8%	−12.0%	−7.3%
SimpleQA Verified	2024-10	40.5%	39.8%	46.0%	14.2%	12.3%	−5.3%	−5.3%
SWE-Bench Verified	2024-05	27.5%	25.1%	28.4%	2.7%	7.6%	1.5%	1.7%

All percentages are estimates of the average quarterly drop in the cost of a given level of accuracy on a given benchmark, over the years of available data. The bottom row of the first panel reports simple averages.. The "non-parametric" values are averages over 100×100 grids that span the benchmark's release date and accuracy range, the latter clipped within the state of the art (SOTA) for that benchmark at each time point. At each grid point, the lowest cost of at-least-as-good performance one quarter later is found and divided by the initial cost. The geometric mean of the ratios is taken and subtracted from 1. All linear, parametric results are coefficients on release year as an explanator for log cost, again reexpressed quarterly. For the Box-Tidwell regressions, the statistic is computed as in the non-parametric column, using model-implied instantaneous decline rates at the grid points. "Model frontier" means modeling the empirical frontier as realized at a grid of points. "Model frontier, require envelopment" means the same, but with the constraint that the fitted surface is never above any data point. "Model all data" means modeling all runs, not just the frontier, with OLS.

Variation in the rate of decline

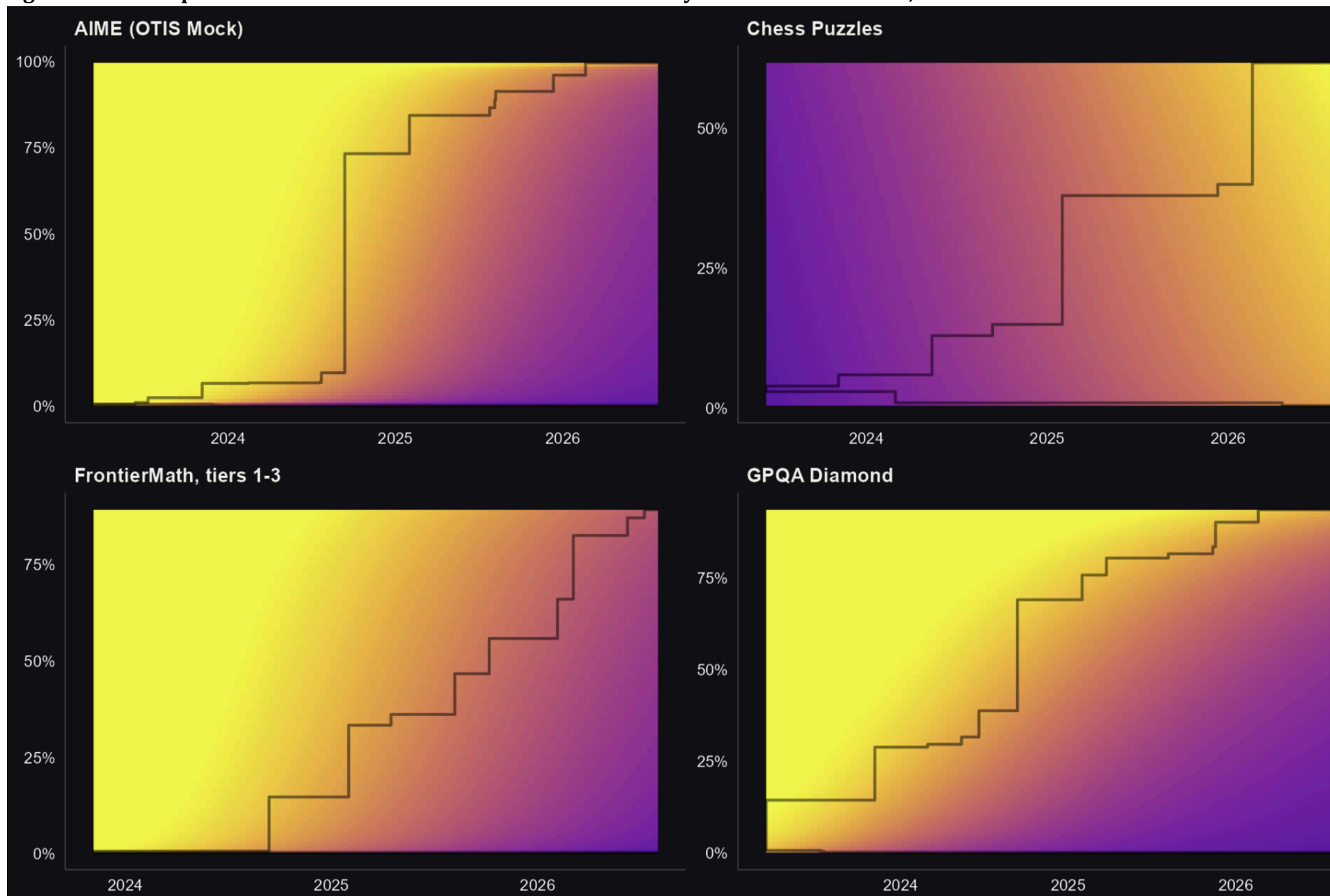
If you take another glance at Figures 2 and 3 and peruse the larger array of plots on the [GitHub page](#), you will see the diversity across benchmarks in the relationship of cost to time and accuracy. Some of the apparent diversity might just be an artifact of overfitting a flexible data to a sample that is too narrow in the time or accuracy dimension for reliable identification of second-order patterns. But some of the diversity is plausibly real, which suggests that the pattern of cost decline does depend on the skill being measured.

Consider Figure 4, which is a screenshot of cost decline heat maps for four benchmarks. Yellow indicates the fastest declines and purple the slowest. To give a sense of where the model is extrapolating beyond the data, staircases are overlaid on the plot to show the strongest and weakest performance yet realized by any model. For three of the benchmarks—AIME; FrontierMath tiers, 1–3; and GPQA Diamond—it appears that the cost declines fastest near state-of-the-art performance. One scenario that would explain this pattern: when a new, SOTA-defining model appears, its maker can briefly charge a premium for this performance. But competitors open and closed quickly catch up, and the price of such performance plunges. Eventually this dynamic fades and the decline slows, even in percentage terms.

Table 2 puts numbers on this pattern. For each benchmark, it reports all the parameter estimates under the Box-Tidwell cost frontier model. Beneath, it lists the model-implied, instantaneous rate of cost decline right when a given performance level debuts as SOTA, and then the same for 1, 2, 3..., up to 8 quarters later. The moment of SOTA debut is estimated from an auxiliary regression of logit accuracy on release date. And, once more, since the model is nonlinear, the instantaneous decline rate depends on the accuracy level. So the displayed statistics are in fact averages over a uniform accuracy grid clipped to the observed accuracy range for the benchmark.

In 3 of the 5 primary benchmarks, and all of the secondary ones, the rate of cost decline falls after SOTA debut. In the exceptions—Chess Puzzles and Mystery Game Puzzles—the rise is small enough that we might view it informally as statistically insignificant. For the primary benchmarks, the decline rate drops from 66%/quarter on average at SOTA to 32% two years later.

Figure 4. Heat maps of cost decline rate vs. release date and accuracy for four benchmarks, Box-Tidwell model of cost frontier



Black staircases show upper bounds on observed performance as a function of time, and in a couple of cases the lower bounds too.

Table 2. Box-Tidwell models for cost frontier

	Primary benchmarks						Secondary benchmarks					
	AIME (OTIS Mock)	Chess Puzzles	FrontierMat h, tiers 1-3	GPQA Diamond	Mystery Game Puzzles	Average	FrontierMat h 2025-02	Frontier Mat, tier 4 h, tier 4 2025-07	Math, tier 4 v2	MATH, level 5	SimpleQA Verified	SWE-Bench Verified
Intercept	154	3.44	86.5	37.4	9.65		51340	7.06	30.1	39.4	31299	11.8
BC accuracy odds	43.9	4.61	16.7	26.3	5.08		19082	0.808	11.5	16.5	14609	2.26
BC time	-496	-0.258	-173	-88.1	-0.950		-390252	-1.56	-8.22	-116	-229944	-18.9
BC accuracy odds × BC time	-132	-0.039	-28.6	-47.7	0.597		-145024	-0.044	-2.80	-36.5	-107297	-0.972
$\lambda_{\text{accuracy}}$	0.237	0.456	0.216	0.64	0.415		0.398	-0.445	0.159	0.423	0.477	0.28
λ_{time}	-3.03	2.73	-1.84	-1.77	1.04		-7.60	0.957		-2.27	-7.34	-1.05
cost drop, %/quarter												
at SOTA	83.3	41.8	63.6	80.7	35.9	66.3	77.7	28.4	85.3	84	83.6	32.9
1 quarter past SOTA	71.1	43.9	57.1	71.5	36.4	58.3	54.2	27.8	79.6	73	62.9	29.4
2 quarters past SOTA	59.3	44.2	51.5	62.5	37	51.8	38.7	27	68.6	62.4	46.1	27.1
3 quarters past SOTA	48.7	44.3	46.6	54.1	37.6	46.6	28.9	26.2	51.1	52.8	33.4	25.2
4 quarters past SOTA	40	44.5	42.3	46.5	38.2	42.4	21.6	25.2	29.6	44.7	24.5	23.5
5 quarters past SOTA	32.8	44.7	38.6	39.9	38.7	39.1	16.3	24.2		37.9	18.1	22.1
6 quarters past SOTA	27	45.1	35.5	34	39.1	36.4	12.1	23.2		32.2	13.3	20.8
7 quarters past SOTA	22.3	45.4	32.5	28.9	39.5	34.2	9.2			27.6		19.6
8 quarters past SOTA	18.3	45.6	29.8	24.6	40	32.4				23.7		18.6
N	4882	3735	2904	5758	4164		2716	1343	1287	4256	3452	4121

Appendix

Stochastic frontier analysis models

This section describes a closely related pair of models that were tried and ultimately set aside.

In economics, one established approach to modeling production and cost frontiers is stochastic frontier analysis (SFA). While in our case there is only one input to production—money—usually SFA models have multiple inputs, like capital and labor, whose supply is jointly bound by a budget constraint. For a given budget, the production frontier is the outer boundary on the domain of possible output levels across affordable mixes of inputs. Most real farms and firms will land inside the boundary, because of inefficiency relative to the technological limit at any given time.

In SFA statistical models, output is a function of input(s), plus *two* error terms. One is conventional and symmetric: it captures random factors in the output variable, such as the output of firms just being high or low by chance, or being mismeasured as such. The other error term is distinctive and asymmetric: it represents the inefficiency of individual producers. Since it enters production functions with a negative sign (or cost functions with a positive sign), it is itself always non-negative. One firm, for example, could have an inefficiency of 10% and another 20%, but none would have an inefficiency of –20%.

An SFA production model looks like

$$y = x'\beta + v_i - u_i$$

The symmetric error term, v_i , is assumed to be normally distributed around 0 with variance σ_v^2 . The asymmetric inefficiency term, u_i , can take one of several forms; here it is assumed to be independently distributed as the positive half of a normal distribution with variance σ_u^2 .

The mental model for the SFA production specification is that there is some frontier representing the limits of the possible, and while some firms lie outside it because of flukes in measurement or random events at the farm in a particular month, many more are on the interior side. And because they are striving for efficiency, as one approaches the frontier on the interior side, the density of firms rises. This picture implies that there is a clear but sharply truncated ridge in the data, where firms bunch against the frontier. If it's strongly

present, then model fitting can detect and separate the two error terms, i.e., the two causes of deviation from the theoretical frontier.

Within the SFA tradition, our modeling context is unusual in that output—performance on a benchmark—is bounded above. So we estimate a bounded variant of SFA:

$$y_i = \sigma(\theta_i + v_i) \text{ with } \theta_i = x_i' \beta - u_i$$

Expressing this as an objective function (log likelihood) to be maximized requires integrating out u_i . If each firm's inefficiency, u_i , were known rather than uncertain and random, then the u_i could be held constant during model fitting and this would be a standard logistic model. It would be fit by finding β to maximize the sum of the log likelihoods of all observations. That is, it would maximize

$$\ln \mathcal{L}|u = \sum_i [y_i \ln \theta_i + (1 - y_{ri}) \ln(1 - \theta_{ri})]$$

where $\ln \mathcal{L}|u$ means the log likelihood when taking all the u_i as given.⁶ Since the u_i are in fact assumed to be independently distributed according to the half-normal distribution with variance σ_u^2 , the log likelihood *unconditional* on them is the expected value of the above over the possible values of the u_i :

$$\ln \mathcal{L} = \sum_i \ln \int_0^\infty du_i f(u_i; \sigma_u^2) \exp[y_i \ln \theta_i + (1 - y_{ri}) \ln(1 - \theta_{ri})]$$

where $f(\cdot; \sigma_u^2)$ is the probability density for u_i under the half normal.

In fact, that objective is not quite the one maximized. In an attempt to stabilize the model fits, u_i is assumed to be the same for each *run*, meaning a set of data points all synthesized from a full run by simulating various budget caps, per the CAISI methodology. So, indexing runs by r and individual observations by r, i , the precise log likelihood is

$$\ln \mathcal{L} = \sum_r \ln \int_0^\infty du_r f(u_r; \sigma_u^2) \exp\left(\sum_i [y_{ri} \ln \theta_{ri} + (1 - y_{ri}) \ln(1 - \theta_{ri})]\right)$$

⁶ Technically, this moves from assuming v_i as normal to assuming it has the logistic cumulative distribution function. The normal-based model, the fractional probit, has a clearer theoretical foundation since by the Central Limit Theorem, normal distributions naturally arise as mixtures of partially uncorrelated random processes. But the logistic distribution is widely used in this context, and in practice the distinction rarely matters.

The integrals are estimated using (non-adaptive) quadrature. To improve precision and speed of the iterative search for the optimal parameters, we had Claude write code to compute, at each trial set of parameter values, the gradient of $\ln \mathcal{L}$ with respect to all of the parameters.

Perhaps not surprisingly, this cleverness ultimately did not produce convincing results: see the first pair of modeling results columns in Table A-1. While the model fitting might theoretically be able to detect a moving frontier, in practice it seems that the frontiers are moving too fast, and the distribution of benchmark tests in their wake too sparse, for the model to reliably distinguish the two error components. For example, for the chess puzzles benchmark, the model finds almost no progress over time, which looks like a mathematically degenerate result. Other estimates are negative.

One variant of the base SFA model is also tried. It is inspired by the visualization that as the production frontier races across time-cost space, the distribution of *available* models, which includes all older models, is increasingly dispersed behind it. So the model allows σ_u^2 to depend on release date. In particular $\ln \sigma_u^2$ is linear in a constant and t . This change does not improve matters: see the relevant columns in the same tables.

Accuracy models

As described in the main text, it is possible to extract implications for the relationship between cost and time even when neither is the dependent variable in the models. The final two pairs of columns in Table A-1 report results based on logistic models of the accuracy frontiers, without and with the envelopment constraint. The results look reasonable. But especially those from the linear models are higher than the model-free averages, as well as the corresponding estimates from the cost models.

Table A-1. Average quarterly rate of cost decline, holding accuracy fixed, dispreferred models

Benchmark	Data start	Model-free	Cost models: Stochastic frontier analysis				Accuracy models			
			Base		Time-varying inefficiency spread		Model frontier		Model frontier, require envelopment	
			Linear	Box-Tidwell	Linear	Box-Tidwell	Linear	Box-Tidwell	Linear	Box-Tidwell
<i>Model order</i>										
Primary benchmarks										
AIME (OTIS Mock)	2023-03	47.1%	−9.6%	−9.8%	−9.9%	−10.4%	68.4%	51.7%	75.8%	70.1%
Chess Puzzles	2023-06	43.0%	6.7%	5.8%	8.1%	6.7%	57.4%	32.5%	52.0%	30.2%
FrontierMath, tiers 1-3	2023-11	53.1%	26.8%	25.4%	27.0%	34.5%	71.5%	71.5%	71.1%	61.5%
GPQA Diamond	2023-03	47.0%	−10.2%	−10.6%	−10.4%	−10.7%	61.0%	45.2%	79.9%	63.7%
Mystery Game Puzzles	2023-06	44.0%	−8.9%	−10.9%	−9.3%	−11.0%	55.3%	55.9%	48.5%	45.8%
Average		47.0%	2.1%	1.1%	2.3%	3.7%	63.3%	53.2%	67.8%	56.4%
Secondary benchmarks										
FrontierMath 2025-02	2024-05	43.3%	−54.4%	−95.3%	−143.0%	−10.0%	57.2%	53.2%	60.5%	48.6%
FrontierMath, tier 4 (2025-07)	2024-06	26.0%	−231.4%	−30.6%	−361.4%	−317.8%	64.8%	59.4%	50.3%	78.2%
FrontierMath, tier 4 v2	2025-01	65.1%	26.6%	32.0%	26.7%	32.2%	78.7%	21.0%	84.3%	49.1%
MATH, level 5	2023-06	51.8%	−22.1%	−3.8%	−108.7%	−79.2%	70.3%	59.0%	81.8%	56.0%
SimpleQA Verified	2024-10	40.5%	−30.2%	−5.1%	−40.1%	−5.1%	60.6%	17.3%	99.0%	17.6%
SWE-Bench Verified	2024-05	27.5%	−3.1%		−3.7%	−3.7%	35.9%	21.0%	40.3%	16.5%

See notes to Table 1. In the accuracy model columns accuracy is the dependent variable and log cost as an explanatory variable, so the decline rates are extracted as $-\beta_t/\beta_{\text{inc}}$. Stochastic frontier analysis models the frontier and the distribution of runs around it with a half-normal distribution of inefficiency.